

MTech Research

Comprehensive Literature Review

Report

Lithium-Ion Battery Systems: Digital Twins · BMS · Multi-Cell Modelling
State Estimation · Prognostics · Hardware-in-the-Loop Validation

Prepared: March 2026

Domain: Electric Vehicle Battery Technology | Power Electronics | AI for Energy Systems

Suitable For: MTech Thesis · IEEE Research Paper · Patent Prior-Art Analysis

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SHEET 1

Digital Twin Technology and Intelligent Battery Management Systems for Lithium-Ion Batteries

Research Theme: Digital Twin for Battery Systems | Papers: 13

PAPER 1 / 13

Comprehensive Analysis on Advances in Battery Modeling, Intelligent Management Systems, and Digital Twin Integration

Literature Review

This paper presents a systematic survey of lithium-ion battery modeling methods and their integration with Digital Twin (DT) technology for Battery Management Systems (BMS) in electric vehicles. The study critically evaluates electrochemical models, equivalent circuit models (ECM), and data-driven approaches, comparing their accuracy and computational costs. The Pseudo-Two-Dimensional (P2D) electrochemical model demonstrates highest accuracy but proves impractical for real-time applications due to computational overhead. A Linear Parameter-Varying (LPV) representation is proposed as a suitable compromise for real-time DT deployment. Key contributions include a proposed DT framework integrated with BMS for dynamic state estimation and predictive maintenance. The study identifies a critical gap in parameter characterization under varying conditions and insufficient data-driven model robustness. Future directions highlight the need for LPV-based DT models and advanced cloud-edge computing integration to enable scalable, real-time battery intelligence in next-generation electric vehicle systems.

PAPER 2 / 13

Enhanced Cloud-Based Digital Twin State Monitoring for Smart Stationary Battery Energy Storage Systems

Literature Review

This research proposes a cloud-based Digital Twin (DT) framework for stationary Battery Energy Storage Systems (BESS) to overcome limitations of purely historical data-driven models. The framework integrates real-time operational data with a virtual model built using Bayesian machine learning, specifically Gaussian Process regression. Validated on a 50 kWh BESS, the DT achieves state estimation deviation of less than 2% during charging cycles, evaluated via RMSE and MSE metrics. The real-time data synchronization significantly enhances predictive maintenance scheduling and operational reliability. A primary limitation is error accumulation during extended charging periods, attributed to model drift in non-stationary operating regimes. The research identifies a critical gap in existing DT frameworks that rely exclusively on historical data, rendering them unreliable under novel real-world conditions. Future work proposes integrating advanced real-time data synchronization protocols and exploring alternative adaptive learning algorithms to improve model fidelity under dynamic load profiles.

PAPER 3 / 13

Overview of Digital Twin Methods Applied to Lithium-Ion Battery Management Systems

Literature Review

This survey examines advancements in lithium-ion battery modeling and the integration of intelligent BMS with Digital Twin (DT) technology across electric vehicle and energy storage applications. The paper reviews electrochemical, circuit-based, and data-driven modeling paradigms, comparing advantages and application suitability. It emphasizes that SOC, SOH, and RUL estimation remain critical performance metrics, yet existing approaches face challenges in balancing model complexity with real-time computational feasibility. A major limitation identified is the high data dependency of data-driven models, where model accuracy degrades significantly under sparse or noisy input conditions. The study highlights the absence of a unified DT framework definition as a structural research gap impeding cross-domain standardization. The paper advocates for convergence of AI, IoT, and cloud technologies within DT-BMS architectures, recommending future research to address standardization, real-time adaptability, and multi-chemistry battery generalization for broader deployment viability.

PAPER 4 / 13

IoT-Enabled Battery Health Management System Based on Digital Twin Technology

Literature Review

This study presents a novel Battery Health Management System (BHMS) that leverages Digital Twin Technology and IoT for real-time predictive monitoring of lithium-ion batteries. The system integrates a Raspberry Pi microcontroller with voltage, current, and temperature sensors connected to a cloud analytics platform implementing threshold-based fault protection algorithms. Experimental validation confirms 93.7% measurement accuracy with a 380 ms fault response time. SOH and RUL prediction employing machine learning models achieve 98.3% fault detection accuracy, extending battery lifespan by 28% over conventional reactive monitoring methods. A significant limitation is the absence of standardized protocols for hybrid renewable-grid-battery integration, limiting large-scale deployment. Research gaps include insufficient uniformity across battery brands and the lack of ML-based aging prediction capabilities in deployed field systems. Future enhancements include blockchain-secured data transmission, adaptive ML degradation recognition, and expansion to multi-chemistry battery configurations for broader industrial applicability.

PAPER 5 / 13

Digital Twin-Based Battery Health Management System for Power Communication Networks

Literature Review

This paper introduces a battery health management system based on digital twin technology, designed for power communication networks to unify and digitize multi-brand battery monitoring. The proposed five-dimensional DT module architecture encompasses physical entity, data perception, data governance, functional modeling, and application layers. Long Short-Term Memory (LSTM) neural networks are employed to analyze real-time and historical operational data for battery status evaluation, anomaly detection, and lifecycle prediction. Key contributions include automated alarm generation and reduction of manual maintenance workload through intelligent monitoring. A notable limitation is the absence of explicit performance benchmarking against alternative architectures. Research gaps identified include lack of brand-agnostic standardization in battery monitoring protocols and insufficient real-time model updating mechanisms. Future directions propose extending the DT framework to broader infrastructure monitoring domains and enabling autonomous decision-making capabilities within next-generation power grid management systems.

PAPER 6 / 13

Design and Development of Smart Battery Health Assessment Model Using Digital Twin Framework

Literature Review

This paper develops a smart battery health assessment model within a Digital Twin (DT) framework, integrating Electrochemical Impedance Spectroscopy (EIS) with MATLAB-based real-time diagnostics. Machine learning models—Multi-Layer Perceptron (MLP) and Long Short-Term Memory (LSTM)—are trained on historical cycling data encompassing resistance, capacity, and temperature features for SOH prediction. The MLP achieves 98.8% prediction accuracy, significantly outperforming LSTM due to battery degradation nonlinearity. A web interface facilitates real-time SOH visualization and early fault detection. Key limitations include the computational demands of integrating real-time EIS and challenges in deploying high-fidelity DT models in resource-constrained embedded environments. Critical research gaps include insufficient validation across diverse battery chemistries and the absence of standardized EIS-DT integration protocols. Future enhancements propose ANSYS-based electro-thermal modeling, advanced EIS diagnostics, and cloud-edge computing to enable distributed, scalable intelligent battery health management in complex operational environments.

PAPER 7 / 13

Digital Twin-Empowered Dynamic SOH Prediction Using Extended Kalman Filter for Lithium-Ion Batteries

Literature Review

This study proposes a real-time State of Health (SOH) estimation methodology for lithium-ion batteries using a data-driven Digital Twin model that reconstructs partial discharge sequences into complete discharge profiles via Extended Kalman Filtering (EKF). The reconstructed data feeds a DWT-LSTM (Discrete Wavelet Transform combined with LSTM) model, achieving over 95% SOH estimation accuracy using only 30% of available discharge data. This approach directly addresses the impracticality of obtaining complete charge-discharge cycles in real EV operations. Key contributions include the novel partial-to-complete discharge reconstruction pipeline and the DT-EKF integration enabling accurate battery health tracking under realistic usage patterns. A significant limitation is the intermediate reconstruction step, which introduces additional computational latency. Research gaps include the need for further validation across diverse battery aging profiles and drive cycles. Future work targets direct incomplete-data SOH estimation without reconstruction and extension to multi-cell battery pack configurations.

PAPER 8 / 13

Digital Twin-Based Model for Battery Energy Storage Systems Enabling Accurate SOC Evaluation

Literature Review

This paper proposes a Digital Twin (DT)-based SOC evaluation approach for Battery Energy Storage Systems (BESS) that overcomes accuracy limitations caused by battery nonlinearity, polarization, and temperature fluctuations. The methodology integrates a multi-neural network fusion strategy employing Recurrent Neural Networks (RNN) and Bi-directional LSTM (Bi-LSTM), enhanced through empirical modal decomposition and correlation analysis for feature isolation. The DT framework improves SOC prediction robustness by mitigating dependency on accurate initial conditions. Experimental results confirm reduced error margins and improved real-time deviation detection compared to single-model baselines. A critical limitation is the current exclusion of temperature and additional state variables from the DT model, reducing accuracy under ambient variation. Research gaps include the need for multi-variable state estimation frameworks and improved temperature compensation models. Future extensions target diverse BESS configurations, multi-variable integration, and deployment in grid-scale renewable energy storage systems for enhanced operational safety.

PAPER 9 / 13

Leveraging Digital Twin Technology for Battery Management Systems: A Systematic Case Study Review

Literature Review

This systematic review investigates DT integration in Battery Management Systems using a PRISMA-compliant methodology, analyzing 122 records across academic and industrial sources. The study identifies five enabling technology pillars for DT-BMS: real-time data acquisition, AI-based state estimation, bidirectional virtual-physical communication, cloud integration, and predictive maintenance algorithms. A comprehensive DT-BMS development workflow is synthesized as a reference architecture. The review reveals that most existing implementations lack detailed virtual models capturing full physical system dynamics and continuous bidirectional data flow—a significant gap in current research. Limitations include restriction to literature published before June 2024 and potential English-language publication bias. Future research priorities include developing user-centric DT interfaces, fostering interdisciplinary collaboration between data science and battery engineering, and optimizing real-time AI diagnostic algorithms for deployment in industrial-scale EV battery management applications.

PAPER 10 / 13

Accurate Electric Vehicle Charge Profile Prediction Using Battery Digital Twin

Literature Review

This paper introduces an adaptive Battery Digital Twin (DT) for generating accurate charge profiles for Battery Electric Vehicle (BEV) fleet management, addressing uncertainties in charging behavior and long-term battery aging. The DT employs a 4-D lookup table architecture parametrized by SOC, reference power, temperature, and aging state, enabling fast, low-cost prediction of complete charge curves. The self-calibrating algorithm continuously updates model parameters by comparing predicted and actual charge profiles, achieving charge time prediction errors below 2% in a real fleet case study—a 7% improvement over standard profile baselines. Key limitations include degraded prediction accuracy under extreme input conditions and lower training data frequency relative to real-time operational data. Research gaps include the absence of bidirectional DT communication and insufficient multi-chemistry generalization. Future work proposes in-field implementation, integration of battery temperature variables, and evaluation across diverse battery chemistries and charger configurations to enhance fleet-level energy management capability.

PAPER 11 / 13

Battery State Prediction Using Digital Twin Framework Integrated with Battery Management System

Literature Review

This paper proposes a Digital Twin (DT) framework integrated with a BMS to enhance real-time SOH and SOC prediction for lithium-ion batteries in electric vehicles. The methodology combines Extreme Gradient Boosting (XGBoost) with Extended Kalman Filtering (EKF), leveraging XGBoost to model nonlinear voltage-current-temperature relationships augmented by EKF for stochastic state filtering. The EKF-XGBoost algorithm achieves significantly improved state estimation accuracy over either method applied independently. A key contribution is the identification of temporal discharge time correlation as a critical feature overlooked in prior BMS studies. Limitations include real-time data handling constraints of onboard BMS hardware and sensitivity of model convergence to feature selection quality. Research gaps identified involve insufficient leveraging of temporal feature correlations and limited hierarchical DT architectures in BMS deployments. Future directions include developing hierarchical cloud-based DT structures for continuous model retraining that accommodate progressive battery aging throughout the full operational lifecycle.

PAPER 12 / 13

Real-Time SOH Estimation for Electric Vehicle Lithium-Ion Batteries Using Machine Learning and Digital Twin

Literature Review

This research implements a Digital Twin model integrated with a K-Nearest Neighbors (KNN) machine learning algorithm for real-time State of Health (SOH) and Remaining Useful Life (RUL) tracking of EV lithium-ion batteries. Equivalent Circuit Models (ECMs) are optimized using genetic algorithms and particle swarm optimization to enhance parameter estimation under time-varying operational conditions. The system successfully predicts SOH at degradation levels of 3%, 6%, 9%, and 12%, demonstrating practical applicability across common EV battery aging scenarios. A principal limitation is the inability to access reliable real-time sensor data in all operational contexts, constraining continuous health monitoring. Research gaps include insufficient exploration of temperature effects on SOH prediction and the absence of validation across multi-cell pack configurations. Future work proposes incorporating advanced AI features for enhanced prediction efficiency, extending the ECM optimization framework to diverse battery chemistries and multi-cell architectures.

PAPER 13 / 13

Data-Model Hybrid Drive Digital Twin Modeling Method for Lithium-Ion Battery State Estimation

Literature Review

This study proposes a data-model hybrid Digital Twin (DT) approach that integrates an electrochemical mechanism model with a data-driven correction layer for lithium-ion battery SOH estimation. An Adaptive Particle Swarm Optimization Support Vector Machine Regression (APSO-SVR) algorithm corrects residual errors between P2D model predictions and measured battery behavior. Operating within a three-component twin architecture—physical entity, digital model, and twin data space—the framework achieves an RMSD of 1.9% with maximum voltage prediction errors below 20 mV. Key contributions include combining mechanistic model interpretability with data-driven adaptability to overcome standalone model deficiencies. A limitation is the paper's absence of explicit scalability analysis for multi-cell configurations. Research gaps include insufficient real-time environmental adaptation and limited validation under extreme temperature and degradation conditions. Future work aims to develop safety-aware intelligent management systems extending the hybrid DT framework to diverse multi-chemistry and multi-cell battery applications.

Sheet 1 Summary | Total Papers: 13

Total Papers: 13. These studies collectively advance the integration of Digital Twin (DT) technology with Battery Management Systems (BMS) for lithium-ion batteries across electric vehicle and energy storage applications. Dominant research trends include cloud-IoT architectures for real-time state estimation, hybrid data-model DT approaches, and ML-enhanced predictive maintenance. Methods range from LSTM and DNN frameworks to KNN and APSO-SVR algorithms embedded within DT pipelines. Key gaps persist in standardized DT framework definitions, cross-chemistry validation, real-time high-fidelity deployment, and scalable multi-cell architectures. Future directions emphasize edge-cloud computing integration, adaptive bidirectional DT communication, and physics-informed hybrid models for next-generation intelligent battery management.

SHEET 2

Hardware-in-the-Loop Simulation Frameworks for Battery Management System Validation in Electric Vehicles

Research Theme: Hardware-in-the-Loop (HIL) for Battery Testing | Papers: 3

PAPER 1 / 3

Functional Analysis of Battery Management Systems Using Multi-Cell HIL Simulator with Electro-Thermal Aging Model

Literature Review

This paper presents the design, characterization, and real-time emulation of a lithium-ion battery electro-thermal multi-cell Equivalent Circuit Model (ECM) coupled with an aging model, validated using experimental data from a 40 Ah commercial Kokam pouch cell. The model is converted to C code and deployed on a dSpace Hardware-in-the-Loop (HIL) simulator, enabling comprehensive BMS functional testing including cell voltage accuracy, temperature tracking, cell balancing, and fault insertion. Electrochemical Impedance Spectroscopy (EIS) and pulse characterization support model parameterization. HIL simulation is demonstrated to be safer, more reproducible, and more cost-effective than physical battery testing, particularly for early-stage fault analysis. Limitations include the inability of simplified non-linear ECMs to fully capture complex battery dynamics. Research gaps identified include the absence of universal battery model libraries for multi-chemistry HIL platforms. Future work targets advanced BMS diagnostic algorithm validation, statistical cell variation modeling, and comprehensive fault insertion analysis within standardized HIL testing environments.

PAPER 2 / 3

Functional Assessment of Battery Management System Performance Tested on Hardware-in-the-Loop Simulator

Literature Review

This study develops a Hardware-in-the-Loop (HIL) simulator framework for online functional assessment of Battery Management Systems (BMS) using a 32-cell series emulated battery pack. The research evaluates SOC estimation accuracy, balancing effectiveness, and energy efficiency under variable C-rates and controlled cell capacity scenarios. Results reveal that BMS SOC estimation deviates by up to 19% at initial charge completion, converging below 1% by cycle end—highlighting significant vulnerability during early charging phases. Energy efficiency exceeds 99.98% under controlled balancing conditions. A key insight is that approximately 7% cell capacity variance from nominal values introduces systematic SOC estimation errors unaddressed by current BMS algorithms. Limitations include the uniform 53 Ah cell capacity assumption, which does not reflect real-world pack heterogeneity. Research gaps include insufficient BMS adaptability to cell-to-cell variation and limited discharge-phase SOC compensation strategies. Future work proposes adaptive BMS algorithms specifically designed to compensate for cell variation effects under dynamic discharge profiles.

PAPER 3 / 3

Hardware-in-the-Loop Simulation Technology and Its Application in Electric Vehicle Development

Literature Review

This paper reviews the concept, development process, and current application status of Hardware-in-the-Loop (HIL) simulation technology in electric vehicle (EV) engineering, emphasizing powertrain parameter matching and control strategy validation. The study analyzes domestic and international HIL practices, identifying modular design and CAN bus integration as key enablers of flexible component testing. HIL platforms are presented as critical tools for reducing development cycles and improving component quality prior to vehicle-level integration testing. A significant limitation identified is that existing HIL simulation models are derived primarily from traditional automobile dynamic theories, introducing considerable inaccuracies for electric-specific drivetrain behaviors. Research gaps include the absence of advanced multi-body dynamic EV models in existing HIL platforms and insufficient anti-jamming and environmental compatibility standards for controllers. Future research directions include developing systematic EV-specific HIL matching theories, enhancing hardware quality, and establishing standardized simulation platform methodologies that accurately represent electric vehicle dynamics.

Sheet 2 Summary | Total Papers: 3

Total Papers: 3. This sheet examines HIL simulation frameworks for BMS validation and EV development. The studies demonstrate HIL as a safe, reproducible, and cost-effective alternative to physical battery testing, with electro-thermal multi-cell ECMs and aging models enabling comprehensive BMS emulation. Key research gaps include the absence of universal battery model libraries, oversimplified ECM representations, and insufficient adaptation of traditional automotive HIL frameworks to electric vehicle-specific requirements. Future directions consistently target advanced SOC compensation, standardized multi-chemistry model libraries, and improved EV-specific dynamic simulation frameworks.

SHEET 3

Multi-Cell Battery Pack Modelling, State Estimation, and Cell Balancing Techniques for Electric Vehicle Applications

Research Theme: Multi-Cell Battery Pack Modelling | Papers: 9

PAPER 1 / 9

Bias Correction-Based State-of-Charge Estimation for Multi-Cell Battery Packs Under Varying Working Conditions

Literature Review

This paper proposes an average cell model augmented with a novel bias correction function for accurate State-of-Charge (SoC) estimation in multi-cell battery packs. The bias function accounts for polarization voltage and temperature effects, modeled using Back-Propagation (BP) neural networks via response surface approximation. An Extended Kalman Filter (EKF) is modified to incorporate bias correction, yielding maximum absolute SoC errors below 2% at 25°C and below 3.5% at 10°C and 40°C. The model's uncertainty quantification framework distinguishes model bias from parameter uncertainty, improving consistency across diverse operating conditions. A primary limitation is the bias function's inadequate temperature dependency when evaluated under abrupt current transients. Research gaps include insufficient integration of aging effects and limited evaluation under fast-charge and regenerative braking profiles. Future work aims to incorporate additional aging factors and advanced ML refinements to improve adaptability across diverse battery configurations and dynamic operational environments.

PAPER 2 / 9

Battery Pack Sizing Methodology for Electric Motorcycles: A Graphical Design Approach

Literature Review

This paper introduces a graphical battery pack sizing methodology for electric motorcycles, providing a visual analytics tool allowing engineers to simultaneously evaluate energy, power, volume, and mass trade-offs across various series-parallel cell arrangements. Total pack specifications are demonstrated to scale proportionally with cell count, while performance characteristics are determined by series-parallel configuration ratios. The methodology facilitates informed design decisions from early engineering stages, reducing costly late-stage redesign iterations. A key limitation of the single-cell hypothesis is that production-induced cell variability and uneven current distribution are neglected, yielding potentially optimistic performance predictions. Research gaps include the absence of cost optimization, thermal modeling, and aging effect integration within the sizing framework. Future work proposes real battery pack prototype validation to compare predicted and actual performance, and methodology expansion to incorporate temperature effects and economic constraints for comprehensive engineering design support.

PAPER 3 / 9

Dynamic Behaviour Analysis of Six-Cell Lithium-Ion Battery Management Systems Through Simulation

Literature Review

This research investigates the dynamic behavior of a six-cell lithium-ion BMS through computational simulation, incorporating capacity, voltage limits, and charge-discharge characteristics with integrated balancing and protective mechanisms. Simulation reveals non-uniform thermal and electrical responses among cells, with Cell 6 exhibiting consistently highest voltage and Cells 2-3 demonstrating most dynamic temperature behavior under standard operating profiles. The study underscores the criticality of individualized cell monitoring in multi-cell configurations over aggregate-level approaches. Key limitations include insufficient robustness of current state estimation algorithms under rapidly changing nonlinear operating conditions. Research gaps include challenges in multi-cell scalability, BMS integration with emerging AI and IoT technologies, and the need for physics-informed simulation under extreme temperature and aging scenarios. Future research anticipates advancement toward solid-state BMS technology with AI-driven control strategies and flexible architectures compatible with diverse battery pack topologies.

PAPER 4 / 9

State-of-Charge Estimator for Lithium-Ion Cell Packs in Electric Vehicles Using Extended Kalman Filtering

Literature Review

This paper proposes an adaptive data-driven SoC estimation methodology for lithium-ion battery packs in electric vehicles using Extended Kalman Filtering (EKF) validated through Simscape-MATLAB simulation. The framework implements a filtering strategy to minimize capacity and resistance variations among individual cells, addressing performance inconsistencies from cell-to-cell differences. Multiple estimation approaches—including Coulomb counting, artificial neural networks, fuzzy logic, and EKF—are comparatively analyzed, with EKF identified as optimal for handling inter-cell variation under dynamic cycling. A critical limitation is EKF sensitivity to covariance matrix initialization and battery model accuracy. Research gaps include the insufficient treatment of individual cell behavior heterogeneity in existing pack-level models and challenges in convergence under fast transient conditions. Future work proposes exploring advanced ANN and fuzzy logic algorithms to further minimize inter-cell variation effects and improve SoC estimation robustness across diverse EV operational profiles.

PAPER 5 / 9

Non-Destructive Capacity Consistency Evaluation of Lithium-Ion Battery Packs Using 1-D Magnetic Field Scanning

Literature Review

This paper proposes a non-destructive testing methodology for evaluating cell capacity consistency in lithium-ion battery packs through one-dimensional (1-D) magnetic field scanning. The technique measures unbalanced current distribution arising from capacity inconsistency by scanning the magnetic field generated during operation, using a magnetic shielding cylinder to minimize ambient field interference. Quantitative evaluation is achieved via magnetic field feature extraction combined with k-Nearest Neighbors (k-NN) classification, achieving a maximum diagnostic error of 4.99%. The non-contact, non-destructive nature of the method makes it practically viable for field maintenance assessments. Key limitations include the requirement for magnetic shielding and the inability to perform quantitative capacity analysis during continuous dynamic discharge. Research gaps include the absence of online integration with BMS platforms and insufficient miniaturization for embedded applications. Future work targets sensor array miniaturization, automated scanning systems, and real-time BMS integration for online capacity consistency monitoring in large-scale battery pack applications.

PAPER 6 / 9

Multi-Cell HIL Simulator-Based Functional Validation of Advanced Battery Management Systems

Literature Review

This paper revisits HIL-based BMS functional validation with emphasis on comprehensive electro-thermal modeling of aging-coupled lithium-ion cells for pre-deployment BMS testing. An electro-thermal multi-cell ECM is designed and validated via EIS-parameterized data from a 40 Ah commercial Kokam pouch cell, incorporating SOC and temperature dependencies for realistic multi-cell behavior simulation. The model is emulated on dSpace HIL hardware and subjected to dynamic load profiles from real EV driving cycles. Testing covers cell balancing, fault insertion, temperature tracking, and current measurement accuracy. The Failure Mode and Effects Analysis (FMEA) methodology is demonstrated as a systematic pre-testing tool for identifying BMS vulnerabilities. Limitations include the inability of simplified ECMs to reproduce all battery dynamic behaviors and the absence of multi-chemistry validation datasets. Research gaps include insufficient statistical cell variation representation and lack of universal model libraries. Future work targets improved aging model representations through statistical cell variation distributions and advanced BMS diagnostic routines for combined SOC-SOH estimation.

PAPER 7 / 9

Mathematical Modelling and Active Cell Balancing Circuit Design for Lithium-Ion Battery Packs

Literature Review

This paper presents an active cell balancing circuit for lithium-ion battery packs employing a single inductor-capacitor (LC) energy converter with bidirectional MOSFET switches to transfer surplus energy from higher-voltage cells to lower-voltage cells without energy dissipation. The circuit operates under complementary PWM control in a near-zero current switching mode, achieving zero voltage disparity between two 6400 mAh Li-ion cells with 95% and 66% SOC within 72 minutes. Peak conversion efficiency is demonstrated, and a comparative analysis with passive balancing confirms the active approach's superiority in energy recovery. A key limitation is the increased system complexity from additional electronic components and dependency on precise PWM control. Research gaps include the absence of scalability analysis for large battery pack configurations and insufficient integration with intelligent BMS control strategies. Future research proposes developing more efficient multi-cell balancing circuits with advanced control algorithms and smart BMS integration to optimize battery safety and maximize operational lifetime.

PAPER 8 / 9

Model-Based Health Condition Monitoring for Multi-Cell Series-Connected Battery Packs Using Dual AEKF

Literature Review

This paper proposes a series dual Adaptive Extended Kalman Filter (AEKF) algorithm for real-time health condition monitoring of multi-cell series-connected lithium-ion battery packs, achieving simultaneous estimation of capacity, internal resistance, and SOC at individual cell level. A reference cell is selected based on average capacity and resistance, with deviations between the reference and remaining cells quantifying pack-level health uncertainty. The algorithm achieves capacity and SOC prediction inaccuracies below 1% and voltage prediction errors below 30 mV. A critical contribution is the real-time uncertainty quantification capability, addressing a significant gap in uncertainty-aware battery pack health assessment. Key limitations include the insufficient relationship between capacity and resistance as sole SOH indicators and challenges from manufacturing-induced cell inconsistency. Research gaps include the absence of real-world EV validation and limited multi-state joint estimation capabilities. Future work proposes comprehensive EV validation and enhanced real-time uncertainty monitoring frameworks for battery pack reliability and safety management.

Reconfigurable Multi-Cell Battery Pack with Simultaneous Charging and Discharging for Portable Electronic Devices

Literature Review

This paper introduces a reconfigurable multi-cell battery pack topology capable of simultaneous charging and discharging, specifically addressing cell balancing challenges in portable electronic devices with inherent manufacturing and operational cell variability. The architecture employs controllable switches and regulated charging current to dynamically redistribute charge, avoiding energy dissipation characteristic of passive balancing. The prototype achieves 97% peak conversion efficiency at 0.2 A load current, with cells initialized at 95%, 60%, and 40% SOC reaching equilibrium within 2.5 hours. Key limitations include increased hardware complexity from multiple switching elements and strong dependency on precise switching control. Research gaps include the limited exploration of the topology in large-scale EV and grid storage applications and the absence of thermal management integration. Future research proposes extending to larger multi-cell packs, developing simplified control strategies, and investigating integration with advanced BMS platforms for broader application viability.

Sheet 3 Summary | Total Papers: 9

Total Papers: 9. This sheet covers multi-cell battery pack modeling, SoC estimation, cell balancing, and health monitoring across electric vehicle and portable device applications. Research trends highlight adaptive EKF-based estimators for multi-cell SoC accuracy, non-destructive magnetic field-based consistency diagnostics, and active balancing circuits for energy-efficient equalization. Recurring limitations include single-cell model inadequacy for heterogeneous packs, absence of thermal-aging co-estimation, and insufficient real-world validation. Key gaps involve scalable BMS solutions for large-format packs and real-time uncertainty-aware health assessment frameworks under dynamic operational conditions.

SHEET 4

Data-Driven and Deep Learning Approaches for Remaining Useful Life Prediction of Lithium-Ion Batteries

Research Theme: Remaining Useful Life (RUL) Prediction | Papers: 4

PAPER 1 / 4

EMD-Enhanced Multi-Step Polynomial Ridge Regression for Lithium-Ion Battery Remaining Useful Life Prediction

Literature Review

This paper proposes an Empirical Mode Decomposition-enhanced Multi-Step Polynomial Ridge Regression (EMPRR) for predicting the Remaining Useful Life (RUL) of electric vehicle lithium-ion batteries. EMD provides signal decomposition and noise reduction, followed by Spearman rank correlation-based feature selection and polynomial ridge regression for multi-step RUL forecasting. Validated across 24 batteries from NASA, CALCE, and HNEI datasets, the model achieves MAE values of 6.25, 20.32, and 14.93 cycles respectively. A distinctive advantage is early-prediction capability, reducing time and cost of standard aging experiments. Key limitations include generalization challenges from inter-battery degradation variability and impracticality of obtaining future health indicator data in real deployments. Research gaps include insufficient representation of fast-charging and regenerative braking protocols in validation datasets. Future research targets additional electrochemical and thermal health indicator identification for improved RUL accuracy under complex realistic EV operating protocols.

PAPER 2 / 4

CNN-DNC and CNN-LSTM-DNC Architectures for Enhanced Lithium-Ion Battery Remaining Useful Life Prediction

Literature Review

This study introduces two novel neural architectures—CNN-DNC and CNN-LSTM-DNC—for Remaining Useful Life (RUL) prediction of lithium-ion batteries, leveraging the Differentiable Neural Computer (DNC) for memory-augmented sequence processing. The CNN component extracts local temporal features, while the DNC provides external memory for long-range temporal dependency capture. On a 124-cell lithium-ion battery dataset, CNN-DNC achieves MAE of 80.133 cycles using 95.2% fewer parameters than CNN-LSTM-DNC, representing a 19.1% accuracy improvement with substantially reduced model complexity. A key limitation is that LSTM preprocessing in CNN-LSTM-DNC may constrain DNC's independent long-range learning capability. Research gaps include insufficient testing on larger and more diverse cross-degradation datasets and limited exploration of transfer learning for cross-battery generalization. Future research proposes integrating GRU and Transformer-based architectures with DNC frameworks and evaluating embedded BMS deployment feasibility for real-time EV battery health monitoring.

PAPER 3 / 4

Multi-Dimensional Feature Extraction and Machine Learning for Lithium-Ion Battery Remaining Useful Life Prediction

Literature Review

This paper investigates multi-dimensional feature extraction from multistage fast-charging data for battery RUL prediction using mainstream machine learning algorithms. Data from the MIT-Stanford public dataset is analyzed with Spearman correlation-based feature selection, identifying four high-correlation features including temperature range (correlation above 0.7). Support Vector Regression (SVR), Gradient Boosting Regression (GBR), and Bayesian Ridge Regression (BRR) models are comparatively evaluated, with the optimal model achieving a 9.34% prediction error and average cycle error of 92.56 cycles. Temperature features are confirmed to encode degradation patterns more effectively than purely electrochemical features. A primary limitation is that multi-stage constant-current charging assumptions may not represent diverse real-world charging behaviors. Research gaps include the absence of fusion model exploration and limited validation under complex discharging conditions outside laboratory protocols. Future work proposes investigating additional ML fusion models and improved capacity variance feature engineering for more accurate life prediction under complex operational conditions.

PAPER 4 / 4

Serial Informer Deep Learning Model for End-to-End Lithium-Ion Battery Remaining Useful Life Prediction

Literature Review

This paper proposes the Serial Informer (SI) deep learning model for end-to-end RUL prediction of lithium-ion batteries, adapting the Informer encoder-decoder architecture with a serial self-attention mechanism for efficient long-sequence correlation capture. The ProbSparse attention mechanism reduces computational complexity compared to standard Transformer architectures while maintaining predictive accuracy. Evaluated on NASA datasets (B0005, B0006, B0007, B0018), the model achieves 17.35% RMSE improvement for B0007 and 14.71% MAPE improvement for B0005 over prior methods. A limitation is that the model requires all battery data properties for sequence correlation computation, potentially including noise-contributing features. Higher computational resource demands relative to simpler architectures challenge real-time embedded BMS deployment. Research gaps include insufficient exploration of feature importance analysis and limited cross-battery generalization validation. Future work proposes integrating complementary ML techniques with the Serial Informer framework and expanding its application to non-battery temporal prediction domains.

Sheet 4 Summary | Total Papers: 4

Total Papers: 4. This sheet covers RUL prediction for lithium-ion batteries using machine learning and deep learning methodologies. Research trends center on multi-feature extraction from charge-discharge data, deployment of DNC-enhanced CNNs, Informer-based Transformers, and ridge regression with EMD preprocessing. A recurring challenge is cross-dataset generalization and early-stage prediction capability under real-world conditions. Future directions consistently target incorporating electrochemical-thermal health indicators, reducing computational overhead for embedded deployment, and validating models under realistic fast-charging and regenerative braking protocols representative of diverse EV operational environments.

SHEET 5

Machine Learning and Deep Learning Methods for State of Health Estimation of Lithium-Ion Battery Systems

Research Theme: State of Health (SOH) Estimation | Papers: 8

PAPER 1 / 8

Deep Neural Network-Based State of Health Prediction for Lithium-Ion Batteries in Electric Vehicles

Literature Review

This paper proposes a deep learning methodology for SOH estimation of lithium-ion batteries employing Deep Neural Network (DNN) and Convolutional Neural Network (CNN) architectures trained on voltage, current, and SOC data extracted via a multi-physics battery model. The DNN achieves the lowest MAE of 0.2253 and MSE of 0.0875, outperforming CNN in both accuracy and computational efficiency. The multi-physics model generates rich representative aging datasets under diverse operating conditions. A significant limitation is the high computational cost and memory requirements of LSTM networks, making them impractical for embedded real-time BMS applications. Research gaps include insufficient temperature dependency modeling in SOH estimation and limited validation across battery chemistries beyond the training configuration. Future research proposes developing temperature-integrated aging models, improving DNN architecture efficiency for embedded deployment, and advancing government-industry collaboration on EV battery technology to reduce costs and extend operational range reliability.

PAPER 2 / 8

Machine Learning-Based State of Health Estimation for Lithium-Ion Batteries in Electric Vehicle Applications

Literature Review

This study investigates Feedforward Neural Network (FNN) and Long Short-Term Memory (LSTM) models for SoH estimation of lithium-ion batteries in electric vehicles using NASA battery degradation datasets. The LSTM model achieves RMSE below 0.05, demonstrating acceptable SoH estimation performance under laboratory conditions. A key research gap identified is the insufficient correlation of SoH predictions with dynamically changing real-world usage profiles, including temperature variation and irregular cycling patterns. Limitations include heavy reliance on historical laboratory data that may inadequately represent real-time operational dynamics and the absence of temperature and impedance features as model inputs. Research gaps extend to the limited incorporation of multi-factorial degradation drivers and insufficient EV chemistry diversity in public training datasets. Future work proposes hybrid ML models incorporating multi-factorial features across diverse battery types to enhance SoH estimation generalizability and practical deployment readiness in EV management systems.

PAPER 3 / 8

LSTM-Based Deep Learning Models for Practical Lithium-Ion Battery State of Health Prediction

Literature Review

This paper presents LSTM-based deep learning models for SOH estimation of commercial 18650 lithium-ion cells, evaluating two-layer LSTM, Bidirectional LSTM (BiLSTM), and CNN-LSTM with attention mechanism architectures. Trained on NASA Prognostics Center datasets (B0005, B0006, B0007) with B0018 for validation, the two-layer LSTM exhibits the most stable performance. The attention mechanism improves sensitivity to rapid aging and capacity recovery phases within cycling data. A critical limitation is the constrained generalization when applied to chemistries or cycling conditions absent from training data. Research gaps include data scarcity across diverse battery brands and chemistries and the need for transfer learning techniques to bridge the domain gap between laboratory and field conditions. Future research proposes expanding multi-chemistry and multi-brand datasets, exploring transfer learning for model adaptability, and integrating impedance measurements and detailed temperature data to improve SOH prediction robustness in commercial EV applications.

PAPER 4 / 8

Multi-Feature Extraction and Fusion-Based SOH Estimation for Large-Format Lithium-Ion Batteries Under Uncertain Aging Modes

Literature Review

This paper proposes a SOH estimation method for large-format lithium iron phosphate (LFP) cells deployed in energy storage stations, combining multi-feature extraction from Incremental Capacity (IC) charging curves with an AI fusion model for robust estimation under uncertain environmental conditions. Nine health indicators derived from IC curve morphological features capture diverse aging signatures, yielding SOH estimation errors of approximately $\pm 1.5\%$ and capacity RMSE between 0.5-0.7%. The methodology demonstrates robustness against cell inconsistency and temperature variability. A key research gap is the insufficient investigation of large-capacity battery SOH under uncertain aging modes and environmental variability. Limitations include model sensitivity to interference and the need for multi-chemistry validation beyond the single LFP dataset. Future research will enhance battery life prediction under diverse working conditions following completion of long-term aging tests and improve uncertainty quantification in AI-based SOH estimation for large-scale energy storage management systems.

PAPER 5 / 8

Support Vector Regression-Based State of Health Estimation for Lithium-Ion Battery Systems

Literature Review

This paper investigates Support Vector Regression (SVR) for SoH estimation of lithium-ion batteries in electric vehicles using operational parameters including voltage, current, and temperature as input features. The SVR framework is implemented in Python and evaluated across varying training dataset sizes to assess data efficiency. Absolute error reduces from 7.26 to 0.8 as the training dataset grows, confirming SVR's excellent generalization capability and fast convergence for small-sample nonlinear regression. Key advantages over experimental measurement approaches include avoidance of systematic error accumulation. Limitations include the necessity for careful parameter tuning—specifically penalty and insensitivity coefficients—and reduced accuracy when applied to batteries operating under out-of-distribution conditions. Research gaps include insufficient hybridization with ensemble or deep learning methods and limited real-time BMS integration validation. Future work proposes combining SVR with hybrid ML approaches for improved accuracy and embedding the framework in advanced real-time BMS platforms for continuous SOH monitoring across EV and energy storage applications.

PAPER 6 / 8

Fusion of Liquid Neural Networks and Multi-Head Attention Mechanisms for Battery Pack SOH Estimation

Literature Review

This paper introduces a novel fusion architecture combining Liquid Neural Networks (LNN) with Transformer-style multi-head attention mechanisms for SOH estimation of lithium-ion battery packs. Learnable liquid time constants (LTC) enable dynamic memory adaptation to varying degradation rates, while multi-head attention identifies critical time steps within sliding window sequences. A coverage-averaging mapping strategy transforms window-level predictions into continuous cycle-level SOH estimates. The framework achieves RMSE of 0.0072 and R-squared of 0.9248 on real battery data, demonstrating competitive performance in capturing complex nonlinear temporal degradation patterns. Limitations include high computational resource requirements and complex model training processes. Research gaps include the absence of cross-chemistry validation and insufficient model compression studies for embedded hardware deployment. Future work proposes validating across diverse battery types and operating conditions and exploring model compression techniques to enable resource-constrained BMS hardware deployment.

PAPER 7 / 8

CNN-Attention-LSTM Hybrid Architecture for State of Health Estimation of Lithium-Ion Batteries

Literature Review

This research proposes a hybrid deep learning architecture integrating CNN, attention mechanism, and LSTM for accurate SoH prediction of lithium-ion batteries. Pearson Correlation Coefficient (PCC)-based feature selection and normalization preprocessing improve model training efficiency, while the Differential-Evolution Grey Wolf Optimizer (DEGWO) enables global hyperparameter optimization. The CNN-Attention-LSTM model achieves MAE of 0.0039, MAPE of 0.0059, and RMSE of 0.0042, outperforming standalone RNN, LSTM, and CNN-LSTM baselines. The attention mechanism is confirmed to significantly improve degradation-critical temporal feature identification. Limitations include the computational cost of traditional Kalman filtering for online state estimation and performance discrepancies between laboratory and real driving conditions. Research gaps include insufficient validation under temperature extremes and cross-battery chemistry generalization. Future work proposes integrating larger, more diverse datasets, exploring advanced neural architectures, and establishing real-time BMS integration for continuous monitoring in EV and industrial energy storage deployments.

PAPER 8 / 8

Integrated SOH Estimation and Balancing Strategy for Multi-Cell-to-Multi-Cell Equalizer Systems

Literature Review

This paper proposes an integrated SOH estimation method and cell balancing strategy for lithium-ion battery packs, utilizing real-time voltage differences between cells to derive balancing current expectations. An intelligent optimization algorithm combined with equalizer circuit control performs simultaneous SOH estimation and active equalization within a multi-cell-to-multi-cell (MC2MC) equalizer topology. The approach achieves SOH estimation relative errors within 2%, with low computational burden and cost suitable for real-time BMS. A key contribution is the ability to estimate SOH directly from balancing operational data without offline characterization cycles. A critical limitation is the inability to estimate SOH for all cells when balancing current approaches zero in static pack conditions. The method relies on a single estimation factor, potentially introducing systematic underestimation. Research gaps include the absence of novel MC2MC hardware validation and insufficient multi-chemistry applicability testing. Future research proposes exploring new hardware topologies, integrating additional state parameters, and enhancing SOH estimation methodologies for static battery pack operating scenarios.

Sheet 5 Summary | Total Papers: 8

Total Papers: 8. This sheet covers SOH estimation for lithium-ion batteries using a spectrum of machine learning and deep learning approaches ranging from SVR and DNN to CNN-LSTM with attention, Liquid Neural Networks, and multi-feature IC-curve fusion models. A consistent challenge is generalization across battery chemistries and operating conditions, with data diversity identified as a primary bottleneck. Two papers address SOH in the context of cell balancing and pack-level estimation, reflecting the field's evolution from single-cell to system-level battery health management. Future directions emphasize transfer learning, physics-informed hybrid models, and lightweight architectures for real-time embedded deployment in commercial BMS applications.

SHEET 6

Advanced State of Charge Estimation Techniques: Model-Based, Data-Driven, and Hybrid Approaches for Lithium-Ion Battery Management

Research Theme: State of Charge (SOC) Estimation for Lithium-Ion Batteries | Papers: 16

PAPER 1 / 16

Robust Fuzzy Logic-Based SOC Estimation and Charging Method Optimization for Lithium-Ion Batteries

Literature Review

This paper proposes a fuzzy logic-based SOC estimation technique integrated with an optimized Constant Current-Constant Voltage (CC-CV) charging strategy for lithium-ion batteries. The fuzzy system continuously monitors voltage, current, temperature, and SOC to enhance charging efficiency and reduce overcharging risk under dynamic conditions. MATLAB simulation confirms that intelligent load dispatching stabilizes system performance and maintains continuous power delivery. The methodology also integrates MPPT-based solar charging to optimize renewable energy extraction. A key limitation is the heavy dependence on membership function and rule set design quality for fuzzy accuracy, with validation confined primarily to simulation rather than large-scale physical battery testing. Research gaps include insufficient real-world validation across diverse battery chemistries and the absence of adaptive rule updating mechanisms for battery aging effects. Future research proposes integrating advanced ML algorithms within the fuzzy framework and exploring its applicability to EV and grid-scale renewable energy storage systems.

PAPER 2 / 16

Hybrid CNN-SBLSTM Deep Learning Approach for SOC Estimation Under Various Drive Cycles and Temperatures

Literature Review

This study presents a hybrid CNN-Stacked Bidirectional LSTM (SBLSTM) neural network for SOC estimation of lithium-ion batteries under varying ambient temperatures and diverse drive cycles. The CNN component extracts two-dimensional local features, while the SBLSTM captures bidirectional temporal dependencies for enhanced sequence modeling. Evaluated across LA92, US06, UDDS, and HWFET drive cycles, the model achieves RMSE values below 0.1% and maximum SOC errors below 0.2%. The research specifically addresses the under-explored problem of SOC accuracy under ambient temperature variability. A key limitation is the dependency of model-based SOC approaches on accurate internal parameter characterization. Research gaps include insufficient adaptive learning for battery aging effects and the absence of real-time embedded deployment validation. Future work proposes integrating adaptive learning mechanisms for aging compensation and optimizing the architecture for embedded real-time BMS deployment in commercial EV applications.

PAPER 3 / 16

Tabu Search-Optimized CNN-Transformer Model for Accurate Battery State of Charge Estimation

Literature Review

This paper introduces a Tabu Search-optimized CNN-Transformer (TS-CNN-Transformer) model for SOC estimation in lithium-ion batteries, combining CNN for local feature extraction with Transformer multi-head attention for long-range temporal dependency capture. Tabu Search optimization automates hyperparameter tuning, identifying an optimal configuration achieving MAE below 1.4%, RMSE below 2.2%, and R-squared exceeding 0.991 across nine real-world datasets. The model outperforms LSTM, GRU, and standalone Transformer baselines in estimation accuracy and robustness. A key limitation is the sensitivity of Kalman Filter baseline methods to noise and equivalent circuit model accuracy. Research gaps include insufficient physics-model integration within the deep learning framework and limited evaluation under highly noisy non-stationary operational conditions. Future work proposes integrating physics-based battery models with deep learning architectures to improve generalization and developing lightweight deployment-ready models for resource-constrained BMS hardware environments.

PAPER 4 / 16

Combined Cell-Level SOC and Temperature Estimation for Multi-Cell Lithium-Ion Battery Packs Using EKF

Literature Review

This paper proposes a combined SOC and temperature estimation method for multi-cell lithium-ion battery packs using an Extended Kalman Filter (EKF) with correlated noise and a forgetting factor, based on a coupled electrical-thermal equivalent circuit model. Simultaneous SOC and temperature estimation reduces the number of required temperature sensors while improving estimation accuracy. Results demonstrate approximately 50% reduction in SOC estimation error when thermal coupling is included. Key contributions include the coupled model that accounts for thermal interactions between adjacent cells, absent in most single-cell SOC frameworks. Limitations include significant computational costs when scaling physics-based models to pack level and degraded temperature estimate accuracy with limited sensor availability. Research gaps include insufficient validation of the approach under aggressive thermal gradients and fast-charge scenarios. Future work proposes incorporating advanced thermal models, adaptive filtering techniques, and integration into commercial EV BMS platforms to improve estimation accuracy while reducing hardware sensor requirements.

PAPER 5 / 16

Hybrid LSTM-EKF Architecture for Robust SOC Estimation in Noisy Battery Operating Environments

Literature Review

This paper proposes a hybrid LSTM-EKF architecture for real-time SOC estimation of lithium-ion batteries under measurement noise and environmental disturbances. The LSTM component captures nonlinear battery dynamics and temporal dependencies, while the EKF refines predictions through state estimation uncertainty management. Experimental results confirm that the LSTM-EKF hybrid achieves lower MAE and RMSE than standalone LSTM and EKF, combining the nonlinear prediction capability of LSTM with the noise-filtering advantages of EKF. Key limitations include the requirement for careful dual-layer parameter tuning and increased computational complexity compared to single-method approaches. Research gaps include insufficient analysis of the hybrid framework's behavior under extreme non-Gaussian measurement noise and limited validation across diverse battery aging states. Future work proposes advanced energy forecasting model integration, adaptive control strategy development, and real-time anomaly detection frameworks to further improve energy management efficiency in EV and renewable energy storage applications.

PAPER 6 / 16

High-Accuracy Iterative EKF-Based SOC Estimation for Lithium-Ion Reconfigurable Smart Battery Systems

Literature Review

This paper proposes a high-precision SOC estimation solution for reconfigurable smart lithium-ion battery systems using an Iterative Extended Kalman Filter (IEKF) that incorporates electrical coupling interactions between individual in-pack cells. An equivalent circuit model is developed to represent the multi-cell reconfigurable architecture, explicitly modeling cell interconnection effects neglected by conventional single-cell SOC methods. The IEKF achieves a maximum SOC estimation error of only 1% across all in-pack cells with MAE and RMSE approximately 0.5% and 0.6%, significantly outperforming standard EKF (maximum error exceeding 3%) and UKF methods. A limitation is that manufacturing-induced cell inconsistencies introduce estimation complexity not fully resolved by the current IEKF framework. Research gaps include insufficient accounting for capacity degradation effects across the operational lifecycle and limited dynamic switching configuration exploitation for enhanced cell parameter estimation. Future research proposes leveraging dynamic switching configurations to enhance individual cell parameter estimation accuracy and validating performance under diverse aging and temperature conditions.

PAPER 7 / 16

Hybrid Low-Dimensional Estimator for Minimum and Maximum SOC Tracking in Multi-Cell Lithium-Ion Battery Packs

Literature Review

This paper presents a computationally efficient low-dimensional hybrid estimator for tracking minimum and maximum SOC among cells in multi-cell lithium-ion battery packs, addressing the infeasibility of individual cell observers in large-scale systems. A switching-based observer mechanism selects a single limiting cell at each time step for SOC estimation using equivalent circuit models. The framework provides analytical guarantees of practical exponential stability, ensuring convergence without Zeno behavior under bounded input currents. Simulation results confirm estimator effectiveness for series-connected pack topologies. A key limitation is the current focus on minimum SOC during discharge without explicit maximum SOC estimation during charging phases. Research gaps include insufficient analysis under bounded perturbations and measurement noise, and limited scalability validation for very large multi-cell systems. Future work proposes addressing joint minimum-maximum SOC estimation under perturbation dynamics and developing enhanced algorithms for improved computational efficiency in large multi-cell battery management systems.

PAPER 8 / 16

Hardware Implementation of Fuzzy Logic SOC Estimation and Adaptive CC-CV Charging for Electric Vehicle Batteries

Literature Review

This paper presents a hardware-based system utilizing fuzzy logic for real-time SOC estimation and adaptive charging optimization of lithium-ion batteries in electric vehicle applications. A microcontroller-based platform continuously monitors individual cell voltage and temperature, implementing an adaptive CC-CV strategy that dynamically adjusts charging parameters based on real-time battery states. Hardware validation alongside Proteus simulation confirms consistent voltage readings across four cells and accurate temperature measurements via LM35 sensors, with SOC categorized into three operational levels with corresponding protective actions. A limitation is the requirement for careful fuzzy rule design and increased hardware complexity and cost compared to software-only implementations. Research gaps include the absence of adaptive rule updating for battery aging and limited validation under multi-chemistry battery configurations. Future work proposes integrating advanced ML techniques with fuzzy inference and deploying the system in smart BMS platforms for high-performance EV and renewable energy storage applications.

PAPER 9 / 16

Fuzzy Logic-Based Robust SOC Estimation and Charging Optimization for Lithium-Ion Battery Packs

Literature Review

This paper investigates fuzzy logic as an adaptive framework for SOC estimation and charging optimization of lithium-ion battery packs, addressing limitations of conventional model-based and Coulomb counting methods under nonlinear behavior and temperature variation. A fuzzy logic controller dynamically adjusts charging rates based on voltage, current, temperature, and SOC feedback, accommodating load demand variation, cell aging, and environmental factors without manual intervention. The simplified rule-based control strategy offers easier implementation compared to PID control approaches, confirming improved SOC accuracy, reduced overcharging and overheating risks, and enhanced battery longevity. Key limitations include the inability of Coulomb Counting baselines to account for aging and temperature and the persistence of nonlinear challenges under extreme variability. Research gaps include the absence of IoT and smart grid integration in current deployments. Future work proposes combining fuzzy logic with ML-based adaptive strategies and smart grid connectivity for real-time monitoring and predictive maintenance to optimize charging rates and extend operational battery lifespan.

PAPER 10 / 16

OCV-SOC-Temperature Relationship Construction and Multi-Branch SOC Estimation for Series-Parallel Battery Packs

Literature Review

This paper proposes a methodology for constructing the Open Circuit Voltage (OCV)-SOC-Temperature relationship for series-parallel connected lithium-ion battery packs, addressing cell inconsistency across temperatures from 5°C to 40°C. Cosine similarity analysis identifies representative cells, and a multi-branch fusion estimation method achieves pack-level SOC estimation. The constructed OCV-SOC-Temperature relationship achieves RMSE of 10 mV at 25°C and within 31 mV at 5°C, significantly outperforming terminal voltage-based approaches. A critical limitation is the absence of capacity aging test data, meaning the proposed approach does not account for SOC estimation accuracy degradation over the battery pack's operational lifetime. Research gaps include the absence of aging effect integration and insufficient validation for parallel-only and mixed configuration topologies. Future work proposes capacity aging tests, extended validation across diverse pack configurations, and refinement for heterogeneous aging-state packs in next-generation BMS applications.

PAPER 11 / 16

AMLRLS and MIAEKF Algorithm-Based SOC Estimation for Lithium-Ion Battery Charging and Discharging

Literature Review

This paper proposes a novel SOC estimation method combining the Adaptive Multi-parameter Linear Recursive Least Squares (AMLRLS) algorithm for online model parameter identification with a Modified Improved Adaptive Extended Kalman Filter (MIAEKF) for real-time SOC estimation. The AMLRLS algorithm reduces model output voltage RMSE by 11.38% compared to conventional FFRLS. The MIAEKF achieves SOC errors within $\pm 2\%$ under both charging and discharging conditions, reducing RMSE by 56% during charging and 79.73% during discharging relative to standard EKF. Key contributions address time-consuming parameter identification and low-accuracy issues in distributed energy storage systems. Limitations include dependency on accurate equivalent circuit model selection and higher computational resource requirements for real-time implementation. Research gaps include limited validation across chemistries beyond the single cell model tested and insufficient extreme temperature evaluation. Future work proposes applying the methodology to large-scale energy storage and EV batteries and exploring further adaptive algorithm refinements for improved estimation under diverse aging and environmental conditions.

PAPER 12 / 16

Informer-LSTM Hybrid Neural Network for Enhanced State of Charge Estimation in Battery Management Systems

Literature Review

This paper presents a hybrid Informer-LSTM neural network for SOC estimation in battery management systems, combining Informer's ProbSparse self-attention for efficient long-sequence feature extraction with LSTM's sequential temporal dependency capture. The hybrid architecture achieves a maximum MAE of 1.395% and maximum RMSE of 1.593%, significantly outperforming pure LSTM baselines. An enhanced Informer decoder minimizes the influence of irrelevant historical data on current predictions, improving estimation stability under varying battery states. A critical limitation is that standard Transformer attention mechanisms suffer reduced accuracy with longer prediction sequences—directly addressed by the Informer's probabilistic sparse attention. Research gaps include insufficient cross-chemistry validation and limited lightweight architecture development for embedded deployment. Future work proposes integrating the model with advanced ML techniques, conducting cross-chemistry robustness validation, and developing deployment-ready lightweight versions for real-time BMS integration in commercial EV and stationary energy storage systems.

PAPER 13 / 16

Improved Unscented Kalman Filter with Weight Optimization for Lithium Battery State of Charge Estimation

Literature Review

This paper presents a joint parameter identification and SOC estimation framework combining Forgetting Factor Recursive Least Squares (FFRLS) with a Weight-Optimized Unscented Kalman Filter (WOUKF) for lithium-ion batteries modeled via a Thevenin equivalent circuit. The WOUKF replaces conventional UKF through optimized Sigma point weight selection to reduce linearization errors. The Recursive Restricted Total Least Squares (RRTLS) method resolves identification bias inherent in standard recursive least squares, improving model parameter accuracy. Experimental results confirm significant accuracy and convergence speed advantages over conventional EKF and UKF methods. Limitations include computational overhead from Jacobi matrix calculations in EKF and dependency on offline model parameters that may not reflect real-time battery aging. Research gaps include insufficient real-time adaptive parameter identification for aging effects and limited evaluation under extreme operating temperature ranges. Future research proposes developing adaptive parameter identification techniques for aging effects and deploying the algorithm in embedded EV BMS hardware for real-time operational validation.

PAPER 14 / 16

Minimum Error Entropy Square Root Cubature Kalman Filter for SOC Estimation Under Non-Gaussian Noise

Literature Review

This paper proposes a Minimum Error Entropy Square Root Cubature Kalman Filter (MEE-SRCKF) for SOC estimation of lithium-ion batteries under non-Gaussian impulsive measurement noise conditions. The MEE criterion replaces conventional Minimum Mean Square Error (MMSE) optimization to provide robustness against heavy-tailed noise distributions. Validated through pulsed discharge and UDDS cycle tests, the MEE-SRCKF achieves RMSE of 0.0112 versus 0.0270 for standard SRCKF, converging rapidly from diverse initial SOC estimates. A key advantage is independence from accurate initial SOC values. Limitations include increased computational complexity from entropy calculations and deployment challenges on low-power embedded systems. Research gaps include the absence of real-world validation under complex multi-modal non-Gaussian noise profiles and insufficient evaluation under dynamic temperature variation. Future work proposes applying MEE-SRCKF to diverse real-world scenarios and investigating enhanced entropy-based optimization criteria to further improve SOC estimation robustness under varying operational and environmental conditions.

PAPER 15 / 16

Guided Ultrasonic Wave-Based SOC Estimation Using GRU Encoder-Decoder Network for Lithium-Ion Batteries

Literature Review

This study introduces a novel SOC estimation methodology using Guided Ultrasonic Wave (GUW) signal features as inputs to a Gated Recurrent Unit-based Encoder-Decoder (GRU-ED) neural network for lithium-ion battery management. GUW signals provide non-invasive access to internal battery state changes including volume, stress, and structural variations imperceptible through conventional voltage-current measurements. The Matching Pursuit (MP) algorithm with a custom asymmetrically dispersive wave dictionary enables sparse signal decomposition and feature extraction. The GRU-ED model achieves MAE of 0.280% and RMSE of 0.377% on a commercial LFP pouch cell, with maximum SOC estimation error of 1.458%, significantly outperforming electrical-signal-based estimators. Key limitations include unexplored temperature and SOH effects on GUW signal characteristics and challenges in feature extraction from multilayered battery structures. Research gaps include the absence of GUW integration with BMS platforms and insufficient multi-chemistry validation. Future work proposes investigating temperature and SOH effects on GUW signatures and validating the framework across diverse battery chemistries for practical BMS integration.

PAPER 16 / 16

SOC Estimation for Reconfigurable Lithium-Ion Batteries Based on Nonlinear Switched System and EKF Observer

Literature Review

This paper proposes a SOC estimation framework for reconfigurable lithium-ion battery circuits using a nonlinear switched system model combined with an Extended Kalman Filter (EKF) observer. The reconfigurable architecture introduces switching dynamics that render conventional SOC estimation methods unreliable when balancing circuits are activated, necessitating a specialized switching EKF observer accounting for changing circuit topology. Experimental results demonstrate approximately 2% estimation error with reduced fluctuation, outperforming non-switching EKF and Luenberger observers in accuracy and convergence robustness. The observer successfully tracks SOC with initial estimation errors of up to 10%. A limitation is the re-convergence time required after each topology switch, potentially affecting tracking in rapidly switching configurations. Research gaps include insufficient thermal and aging effect integration in the switching observer model and limited scalability validation for larger multi-cell reconfigurable systems. Future research proposes improving estimation under varying temperature conditions and battery aging and developing advanced Kalman observer architectures to reduce post-switch convergence time.

Sheet 6 Summary | Total Papers: 16

Total Papers: 16. This sheet covers SOC estimation for lithium-ion batteries using the broadest methodological range across all sheets, including fuzzy logic, diverse Kalman filter variants (EKF, UKF, IEKF, AMLRLS-MIAEKF, MEE-SRCKF), deep learning architectures (CNN-SBLSTM, Informer-LSTM, TS-CNN-Transformer), hybrid ML-Kalman frameworks (LSTM-EKF), physics-based OCV-SOC-temperature modeling, and novel GUW-based signal processing. A consistent trend is the convergence of hybrid data-driven and model-based methods for improved accuracy and noise robustness. Key challenges include generalization across aging states, computational efficiency for embedded deployment, and handling non-Gaussian measurement noise in operational environments.

Overall Research Summary

Sheet	Research Category	Theme	Papers
1	Digital Twin Technology and Intelligent Battery Management S...	Digital Twin for Battery Systems	13
2	Hardware-in-the-Loop Simulation Frameworks for Battery Manag...	Hardware-in-the-Loop (HIL) for Battery Testing	3
3	Multi-Cell Battery Pack Modelling, State Estimation, and Cel...	Multi-Cell Battery Pack Modelling	9
4	Data-Driven and Deep Learning Approaches for Remaining Usefu...	Remaining Useful Life (RUL) Prediction	4
5	Machine Learning and Deep Learning Methods for State of Heal...	State of Health (SOH) Estimation	8
6	Advanced State of Charge Estimation Techniques: Model-Based,...	State of Charge (SOC) Estimation for Lithium-Ion Batteries	16
TOTAL	All Research Categories	Lithium-Ion Battery Systems	53

This comprehensive literature review systematically analyzes **53 peer-reviewed research papers** across six thematic categories covering the broad and rapidly advancing domain of Lithium-Ion Battery Systems. The reviewed works collectively illustrate a paradigm shift from traditional reactive battery management toward intelligent, data-driven, and hybrid frameworks that integrate Digital Twin technology, deep learning architectures, and advanced Kalman filtering variants into cohesive Battery Management System (BMS) platforms. **State of Charge estimation** (16 papers) emerges as the most extensively researched topic, underscoring its foundational role in battery management. **Digital Twin integration** (13 papers) represents the most dynamic and rapidly advancing frontier. **Multi-cell pack modeling** (9 papers) and **SOH estimation** (8 papers) address critical system-level engineering challenges, while **RUL prediction** (4 papers) and **HIL simulation** (3 papers) provide targeted solutions for prognostics and validation. Cross-cutting themes include the tension between model accuracy and computational feasibility for real-time embedded deployment, challenges in generalizing models across battery chemistries and aging states, and the growing convergence of physics-based and machine learning approaches. This review provides a comprehensive foundation for MTech thesis development, IEEE paper background sections, and patent prior-art analysis in the domain of intelligent battery management systems.